This is a document to inform wiring, programming, and diagnosis of the Pi 5, sensors, and motors.



Note: Refer to Final_Electronics_Parts_List.xlsx for datasheets.

Pi 5 Complete Pinout

| Primary | Alternate Functions | Pin | Notes: | Pin | Primary | Alternate Functions |
|---|---|---|---|---|---|---|
| **3.3V** | DC Power (Max 50mA total across all 3.3V pins). | **1** | **- Almost every GPIO can be mapped to multiple internal controllers (UARTs 2–5, SPIs, etc.) via the "Device Tree."**<br>- Current Limits: The total current draw across all 3.3V pins should not exceed **50mA**.<br>- Voltage Warning: All GPIO pins use **3.3V logic**. Connecting 5V directly to a GPIO pin will likely fry the RP1 chip. | **2** | **5V** | DC Power (Direct from USB-C input). |
| **GPIO 2** | **I2C1 SDA**; Fixed 1.8kΩ hardware pull-up resistor. | **3** | | **4** | **5V** | DC Power (Direct from USB-C input). |
| **GPIO 3** | **I2C1 SCL**; Fixed 1.8kΩ hardware pull-up resistor. | **5** | | **6** | **Ground** | 0V Reference. |
| **GPIO 4** | **GPCLK0** (General Purpose Clock output). | **7** | | **8** | **GPIO 14** | **UART0 TX** (Primary Serial Console). |
| **Ground** | 0V Reference. | **9** | | **10** | **GPIO 15** | **UART0 RX** (Primary Serial Console). |
| **GPIO 17** | SPI1 CE1 / UART3 RTS. | **11** |  | **12** | **GPIO 18** | **PWM0** / PCM_CLK (Audio Clock). |
| **GPIO 27** | SPI1 SCLK / UART4 RTS. | **13** | | **14** | **Ground** | 0V Reference. |
| **GPIO 22** | SPI1 MISO / UART4 CTS. | **15** | | **16** | **GPIO 23** | SPI1 CE0 / UART4 TX. |
| **3.3V** | DC Power (3.3V Rail). | **17** | | **18** | **GPIO 24** | SPI1 MOSI / UART4 RX. |
| **GPIO 10** | **SPI0 MOSI** / UART5 TX / PWM1 (via Alt). | **19** | | **20** | **Ground** | 0V Reference. |
| **GPIO 9** | **SPI0 MISO** / UART5 RX / PWM1 (via Alt). | **21** | | **22** | **GPIO 25** | SPI1 SCLK / UART2 TX. |
| **GPIO 11** | **SPI0 SCLK** / UART5 RTS. | **23** | | **24** | **GPIO 8** | **SPI0 CE0**. |
| **Ground** | 0V Reference. | **25** | | **26** | **GPIO 7** | **SPI0 CE1**. |
| **GPIO 0** | **ID_SD**; Reserved for HAT ID EEPROM (I2C). | **27** | | **28** | **GPIO 1** | **ID_SC**; Reserved for HAT ID EEPROM (I2C). |
| **GPIO 5** | GPCLK1 / UART3 TX. | **29** | | **30** | **Ground** | 0V Reference. |
| **GPIO 6** | GPCLK2 / UART3 RX. | **31** | | **32** | **GPIO 12** | **PWM0** / UART5 TX. |
| **GPIO 13** | **PWM1** / UART5 CTS. | **33** | | **34** | **Ground** | 0V Reference. |
| **GPIO 19** | **PWM1** / SPI1 MISO / I2S MCLK. | **35** | | **36** | **GPIO 16** | SPI1 CE2 / UART3 CTS. |
| **GPIO 26** | SPI1 CE2 / UART2 RTS. | **37** | | **38** | **GPIO 20** | PCM_DIN (I2S Audio Data In). |
| **Ground** | 0V Reference. | **39** | | **40** | **GPIO 21** | PCM_DOUT (I2S Audio Data Out). |

Task vs GPIO# Reference Table:

| Task | Identification by GPIO # on which it is located. | Operation | ? |
|---|---|---|---|
| Start LED Detection Color Sensor | Through Multiplexer | | |
| Driving | GPIO 17 Left Forward<br>GPIO 27 Left Reverse<br>GPIO 12 Left PWM<br>GPIO 0 Right Forward<br>GPIO 10 Right Reverse<br>GPIO 13 Right PWM<br><br>GPIO 9 Left Encoder1<br>GPIO 11 Left Encoder 2<br>GPIO 21 Right Encoder 1<br>GPIO 26 Right Encoder 2 | | |
| Multiplexer (All Tasks) | GPIO 2 I2C1 SDA<br>GPIO 3 I2C1 SCL | | |
| Button Task | GPIO 5 9g Servo | | |
| Keypad Task | GPIO 16 Limit Switch<br>GPIO 5 9g Servo PWM<br>GPIO 6 9g Servo PWM | | |
| Crank Task | GPIO 16 Limit Switch<br>GPIO 24 15kG Servo PWM | | |
| Color-Sensor Arm | GPIO 4 15kG Servo PWM<br>Color Sensor (Through Multiplexer) | | |
| OLED Display | Through Multiplexer | | |
| Drone Handling | GPIO 24 Holding Servo PWM<br>GPIO 20 Localization IR Sensor | | |
| Duck Handling | Duck ToF Sensor (Through Multiplexer)<br>GPIO 18 Duck Brush PWM<br>GPIO 19 Duck Belt PWM | | |
| **TBD ITEMS** | | | |
| Front ToF Sensor | GPIO ? I2C1 SDA<br>GPIO ? I2C1 SCL | | |
| Back ToF Sensor | GPIO 22 I2C1 SDA | | |

| | | | |
|---|---|---|---|
| | GPIO 23 I2C1 SCL | | |
| Misc Limit Switch | GPIO 1 | | |

Motor/Servo Information for Programming

| What it Is | Data Type | Information | Model # |
|---|---|---|---|
| Drivetrain Motor Driver | Per Motor:<br>3x GPIO<br>2 pins are On/Off Output for each direction.<br>1x PWM for speed (called Enable pin) | High-Speed PWM: It supports up to 60kHz PWM, allowing for silent motor operation.<br><br>Blink Codes: STATE LED is diagnostic tool:<br>Slow Blink: Under-voltage (Battery low).<br>Fast Blink: Over-voltage (Braking spikes).<br>Double Blink: Overheat (Output disabled). | DF ROBOT DFR0601 |
| Drive Motor Encoders | Per Motor:<br>2x GPIO<br>Quadrature Signals (pulse counting) | Resolution is 1,245.768 counts per revolution (CPR) at output shaft. Because this is not an integer, "distance traveled" calculations will drift over time if you don't use floating-point math for pulse-to-degree conversion.<br><br>Voltage Amplitude: signal output level (HIGH) is equal to whatever you supply to Encoder VCC.<br>Warning: If powering encoder with 5V, must use a voltage divider or level shifter before connecting A/B to Pico's 3.3V GPIOs. | 638298<br>encoder_use_parameter<br><br>116rpm_gearmotor_specifications |
| 180° Micro Servos | PWM maps to<br>> **Position** | PWM: 500 – 2500 μs<br>"Digital" servo but requires 50Hz–333Hz PWM signal. EMI noise; keep signal wires away from motor power.<br>~5μs deadband. If code updates angle too frequently with tiny increments, servo may not move at all until change exceeds this threshold. | MG90S |
| Continuous Servos | PWM maps to<br>> **Speed** | | TD-8815MG |

Sensor Information for Programming

| What it Is | Address | Information | Library(s) | Model # | |
|---|---|---|---|---|---|
| 4 Channel i2c Multiplexer<br><br>2.3-5.5V | I2c,<br>Address:<br>0x70<br>Write:<br>0xE0<br>Read:<br>0xE1 | First, send control byte to PCA9546's 0x70 to indicate desired channel:<br>0x01 Channel 0<br>0x02 Channel 1<br>0x04 Channel 2<br>0x08 Channel 3<br>0x00 None (Reset)<br>Then, send i2c command exactly as if the sensor was plugged directly into Pico using the sensor's own i2c address.<br><br>NOTE: *Can* enable multiple channels at once if sensors have different addresses, or on another SDA/SCL bus. ToF sensors' addresses may be changed during initialization with use of their XSHUT pins.<br><br>Reset Pin: The Multiplexer has a hardware /RESET pin. If your code hangs, toggling this pin low then high will deselect all channels and reset the internal state machine.<br><br>Pull-up Resistors: Pico's internal pull-ups (gpio_pull_up) are often too weak for a mux setup. You should have physical resistors (2.2kΩ to 4.7kΩ) on Master bus (Pico side) and on each sub-bus for reliable 400kHz operation. | | PCA9546 | |
| Color Sensor (Start LED & Antenna Identification)<br><br>2.6V–3.5V | i2c<br>Address:<br>0x29<br>Read:<br>0x52<br>Write:<br>0x53 | Requires integration time (e.g., 2.4ms to 700ms) set in ATIME register before valid data can be read.<br><br>"Slow" Sensor: If you set a high-accuracy integration time (e.g., 614ms), you cannot poll the sensor faster than that. Check "Status Register" (0x13) bit 0 to see if data is actually ready before reading.<br><br>Saturation Quirk: sensor has a 16-bit ADC (max 65,535). If it's too close to a light source or "Integration Time" is set too long, registers will max out. You must tune ATIME (0x81) and CONTROL (0x8F, for gain) together. | | TCS34725 | |

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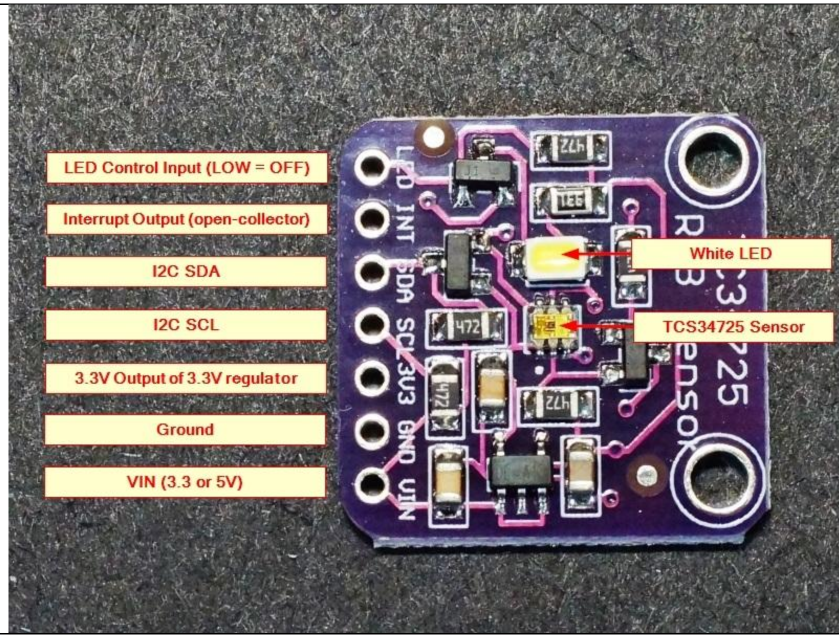
|---|---|---|---|---|---|
| | |  | | | |
| Duck 18° FOV Time-of-Flight Distance Sensor<br><br>2.6V–3.5V | i2c<br>Address:<br>0x29<br>Read:<br>0x52<br>Write:<br>0x53 | | | VL53L4CD | |
| 1” OLED Display Module<br><br>3.3V–5.0V | i2c<br>Address:<br>0x3C or 0x3D<br>Read:<br>0x78<br>Write:<br>0x79 | Resolution is Unknown at moment.<br><br>Charge Pump Command: Most hobby OLEDs don't have an external high-voltage supply. You must send "Charge Pump" command sequence during initialization, or screen will remain black:<br>Send 0x8D (Charge Pump Command)<br>Send 0x14 (Enable Internal Charge Pump)<br><br>Memory Addressing: In Pico SDK, you typically want "Horizontal Addressing Mode" (0x20, then 0x00). Otherwise, data will wrap around incorrectly. | | ER-OLED SSD1306 | |
| Long range Time-of-Flight Distance Sensor<br><br>2.6V–3.5V | 1x SDA<br>1x SCL<br>i2c<br>Address:<br>0x29<br>Read:<br>0x52 | "Boot State" Quirk: After power-on, you must wait for sensor to boot before sending commands. Check FIRMWARE__SYSTEM_STATUS register; it should return 0x03 when ready.<br><br>Alt Function Logic: To move I2C away from the standard pins, have to tell Pi's firmware to remux the pins. For example, if I2C3 on GPIO 22/23, add this to /boot/firmware/config.txt: | | VL53L5CX | |

| | | | | |
|---|---|---|---|---|
| | Write:<br>0x53 | “ # Enable I2C3 on GPIO 22 and 23<br>dtoverlay=i2c3,pins_22_23 ”<br><br>If not going through multiplexer: XSHUT Pin: These sensors share same address. Must wire their XSHUT pins to separate GPIOs on Pico. Pull all XSHUT low, then bring them up one by one to reassign addresses using i2c_SLAVE__DEVICE_ADDRESS register.<br><br>Byte Swapping: Many C implementations of ST Ultra Lite Driver (ULD) have a "Big Endian vs. Little Endian" conflict. If distance readings are massive (e.g., 55,000 mm), code is likely swapping High and Low bytes. | | |
| Limit Switches | 1x GPIO On/Off Input | Use stronger pull-up resistor (~1kΩ to 4.7kΩ) to ensure enough current flows to "clean" contact.<br><br>Mechanical Bounce: Being a mechanical switch, it will "bounce" for 5–20ms. You must implement a software debounce timer or a hardware RC filter to avoid multiple triggers per press.<br><br>Terminal Logic: * COM to NO: Circuit closes when pressed (Standard limit). COM to NC: Circuit opens when pressed (Safety "kill-switch" mode). | | V-156-1C25 |

Appendix (For Acronyms)

**Pin Roles**

GPIO - General Purpose Input/Output. For digital input/output w/ configurable internal pull-up/pull-down resistors.

TX / RX - Transmit / Receive for serial UART communication. Dedicated lines for asynchronous data transmission/reception.

SDA / SCL - Serial Data / Serial Clock for I2C bus. Lines for bidirectional data transfer/synchronization of timing.

MOSI / MISO - Master Out Slave In / Master In Slave Out for SPI bus. Primary data lines for synchronous transfer / modernly referred as COPI/CIPO.

SCLK - Serial Clock for SPI bus. Synchronizing pulse for data bits generated by master device.

CE - Chip Enable for SPI bus. Logic signal for selecting specific peripheral for active communication / also known as Chip Select (CS).

RTS / CTS - Request to Send / Clear to Send for UART. Signals for hardware flow control to prevent data buffer overflow.

GPCLK - General Purpose Clock. Output pins for providing precise square wave frequencies for external hardware/integrated circuits.

ID_SD / ID_SC - I2C pins for identification of HATs. Dedicated bus for reading configuration data of EEPROM on expansion boards for automatic driver loading.

MCLK / DIN / DOUT - Master Clock / Data In / Data Out for I2S/PCM. System clock/serial data lines for movement of digital audio signals.

**Communication Protocols**

PWM - Pulse Width Modulation. Simulates analog signals by pulsing digital signals on/off for control of motors/LEDs.

UART - Universal Asynchronous Receiver-Transmitter. Serial protocol for asynchronous communication between devices using two wires for data exchange w/o shared clock.

I2C - Inter-Integrated Circuit. Synchronous/multi-master/multi-slave bus for controller communication with sensors/displays using unique 7-bit/10-bit addresses.

SPI - Serial Peripheral Interface. High-speed/full-duplex synchronous data bus for master-slave communication.

I2S / PCM - Inter-IC Sound / Pulse Code Modulation. Specialized protocols for transmitting digital audio data via bit-clock/word-select lines.

**General Hardware Terms**

SoC - System on Chip. BCM2712 processor housing CPU/GPU/RAM for primary operations.

RP1 - Southbridge chip of Pi 5. Manages I/O via PCIe 2.0 interface.

DC - Direct Current. Unidirectional flow of electric charge.

Ground (GND) - 0V reference. Common return path.